Problem 1.2

You borrow \$12,000 from a bank. The loan is to be repaid in full in one year's time with a payment due of \$12,780.

1. What is the interest amount paid on the loan?
2. What is the annual interest rate?

Solution

(a) Interest amount

The interest paid is the difference between the amount repaid and the amount borrowed:

$$\text{Interest} = 12{,}780 - 12{,}000 = 780.$$

Therefore, the interest amount paid is

$$\boxed{\$780}.$$

(b) Annual interest rate

The annual interest rate is the interest divided by the principal:

$$r = \frac{12{,}780 - 12{,}000}{12{,}000}.$$

So,

$$r = \frac{780}{12{,}000} = 0.065.$$

Hence, the annual interest rate is

$$\boxed{6.5\%}.$$